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must attempt conclusions and explanation to gain max

allow salt for NaCl **bod** ref. to, salinity / salinity stress (indicates NaCl)

allow either mp2 **or** mp4, and allow manipulated data for these

1 (for, root / shoot, overall main trend is) increasing NaCl concentration, decreases dry mass / growth is more inhibited ;

2 comparison data, any two concentrations, root or shoot dry mass ;

control

NaCl	root	shoot
0 mmol dm ⁻³	0.53 g	0.24 g
110 mmol dm ⁻³	0.14 g	0.09 g
220 mmol dm ⁻³	0.04 g	0.04 g
330 mmol dm ⁻³	0.02 g	0.02 g

*I NaCl conc of zero
MeJa (*0=0.19g and 0.12g)

NaCl	root	shoot
110 mmol dm ⁻³	0.21 g	0.12 g
220 mmol dm ⁻³	0.13 g	0.06 g
330 mmol dm ⁻³	0.06 g	0.04 g

3 idea that MeJA presence produces, better growth / higher (dry) mass , when NaCl present ;

4 comparison data, one concentration, control and MeJA, root or shoot dry mass ;

root shoot

NaCl	control	MeJA
110 mmol dm ⁻³	0.14 g	0.21 g
220 mmol dm ⁻³	0.04 g	0.13 g
330 mmol dm ⁻³	0.02 g	0.06 g

NaCl	control	MeJA
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Question	Answer	Marks
3	5 MeJA presence, inhibits growth / decreases dry mass / results in less growth, with no NaCl / at 0 mmol dm⁻³ ; ora A MeJA presence with, no NaCl / at 0 mmol dm⁻³, in root is 0.19 g and control is 0.53 g / in shoot is 0.12 g and control is 0.24 g <i>max 3 out of 4 for effect of MeJA</i> 6 MeJA, transported / AW, to roots, in phloem sap / by mass flow / in phloem ; 7 target cells / receptors, (in roots) for binding of MeJA ; 8 binding of MeJA, stimulates (protective) response in cells ; <i>context is overall response</i> I <i>transduction / second messengers / enzyme cascade etc.</i> 9 AVP ; suggestion of cell-signalling response that occurs to give tolerance e.g. increased, water uptake / osmosis (into, roots / cells) efflux, sodium ions / chloride ions partitioning ions within vacuoles of root cells less water leaves cells AW enzymes produced to help growth <i>if mp9 gained and MeJA binding stated, then also award mp8</i>	